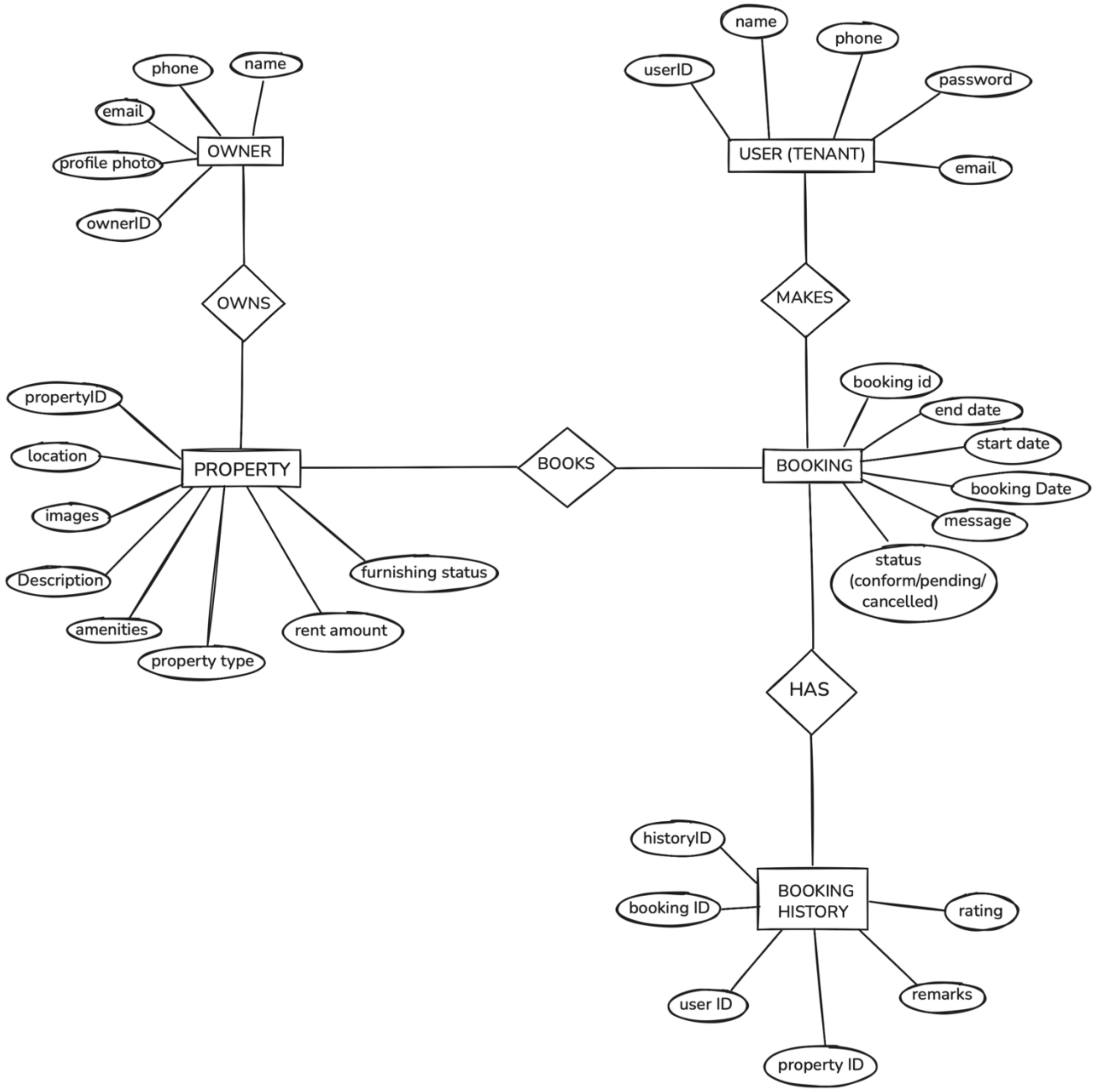
ER Diagram-



Explanation of ER Diagram

The ER Diagram represents the core structure of the **House Rent Application**. The system consists of four main entities: **User, Property, Booking, and Booking History**. Users can search for available rental properties and make bookings, while property owners can list their properties on the platform. The Booking entity manages rental requests, and Booking History maintains records of all booking activities and status updates.

Entities & Attributes

1. User

Attributes:

- UserID (PK)
- Name
- Email
- Phone
- Password
- ProfileImage

Explanation:

The User entity represents tenants who register on the platform to search for rental properties. Users can browse property listings, view details, and make booking requests for properties they are interested in renting.

2. Property

Attributes:

- PropertyID (PK)
- OwnerID
- Title
- Description
- Location
- PropertyType
- RentAmount
- Amenities
- Images
- Status (Available/Booked)

Explanation:

The Property entity stores all information related to rental properties listed on the platform. Each property belongs to an owner and contains details such as location, rent amount, property type, amenities, and availability status.

3. Booking

Attributes:

- BookingID (PK)
- UserID (FK)
- PropertyID (FK)
- BookingDate
- VisitDate
- Status (Pending/Confirmed/Cancelled)
- Message

Explanation:

The Booking entity records booking requests made by users for a property. It contains information about the user, the selected property, booking dates, and the current booking status.

4. Booking History

Attributes:

- HistoryID (PK)
- BookingID (FK)
- UserID (FK)
- PropertyID (FK)
- Action
- ActionDate
- Remarks

Explanation:

The Booking History entity maintains a complete record of booking-related activities. It stores information such as booking confirmations, cancellations, updates, and other actions performed during the booking process.

Relationships Overview

User ↔ Booking

- One user can make multiple bookings (1:N).
- Each booking belongs to one user.

Property ↔ Booking

- One property can receive multiple booking requests (1:N).
- Each booking is associated with one property.

Booking ↔ Booking History

- One booking can have multiple history records (1:N).
- Each history record belongs to one booking.

User ↔ Property

- Users can view and search multiple properties.
- Properties are listed on the platform for users to rent.